# REAL-TIME STOCK MARKET WATCHLIST & ALERT SYSTEM

PROJECT REPORT

Submitted by

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Prepared For

**DATABASE SYSTEMS (BCSE302P) -Lab PROJECT**

Submitted To

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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

# Abstract

In financial markets around the world, the prices of instruments provided by them fluctuate with time. This is not an issue for large organizations or individuals with their careers focused on this, but for a casual individual this creates a huge challenge. The tool - **Real-Time Stock Market Watchlist & Alert System** developed by us aims to solve this problem. By allowing users to create personalized watchlists and alerts, we are providing a solution aimed towards reducing the friction between keeping track of the instruments and actually investing.

Our tool allows users to create multiple watchlists each with its own set of instruments allowing for a personalized monitoring system. The price variation of each instrument is stored as a **price-tick**, containing details such as the instrument, observed time, price, volume, data source allowing for a granular overview of the instrument. The user can also set custom **ABOVE/BELOW alerts** which allows the user to be notified in case of the price ever crossing above or below the threshold. Further all the data - including the events generated or ingested by the user or tool are stored in the database for a **deterministic replay feed** allowing the user to further analyze and learn from the market.

The tool is implemented using **PostgreSQL** as the database. We chose PostgreSQL because the tool uses a highly-relational database schema with multiple entities. PostgreSQL also provides strong support for keys, constraints and indexing allowing for optimized data ingestion and retrieval. PostgreSQL provides reliable transaction management, concurrency control and **ACID** guarantees allowing the tool to monitor and ingest data for hundreds of instruments every second. It also allows complex SQL queries allowing us to efficiently retrieve related data across multiple entities.

The database is accessed by users through a backend developed in Python with FastAPI. The backend acts as the interface between the user and the database, handling operations such as creating watchlists, adding instruments, configuring alerts, ingesting price data, and retrieving historical information. The relational structure of the database, combined with constraints and normalization, helps reduce data redundancy and maintain consistency across these operations.

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# 1. INTRODUCTION

In financial markets around the world, the prices for each instrument catered by them vary every second. This makes it difficult for individuals to keep track of instruments of their interest. This tool allows users to create watchlists through which they can monitor the price changes of instruments, define custom rules with respect to the price of the instrument and be notified in case of an event of their interest. This is achieved by a database-driven monitoring approach. This approach keeps the bulk of the processing on the database side making the entire user experience smooth and seamless.

# 2. PROJECT SCOPE

- Multiple named watchlists per user; add and remove instruments; latest price for each
- Price history with latest-price and history views
- Custom ABOVE/BELOW alert rules - edge-triggered
- Durable alert history for analysing and monitoring instruments
- Offline, deterministic replay feed to simulate movement of instruments
- User interface for seamless interaction

# 3. KEY CONTACTS AND STAKEHOLDERS

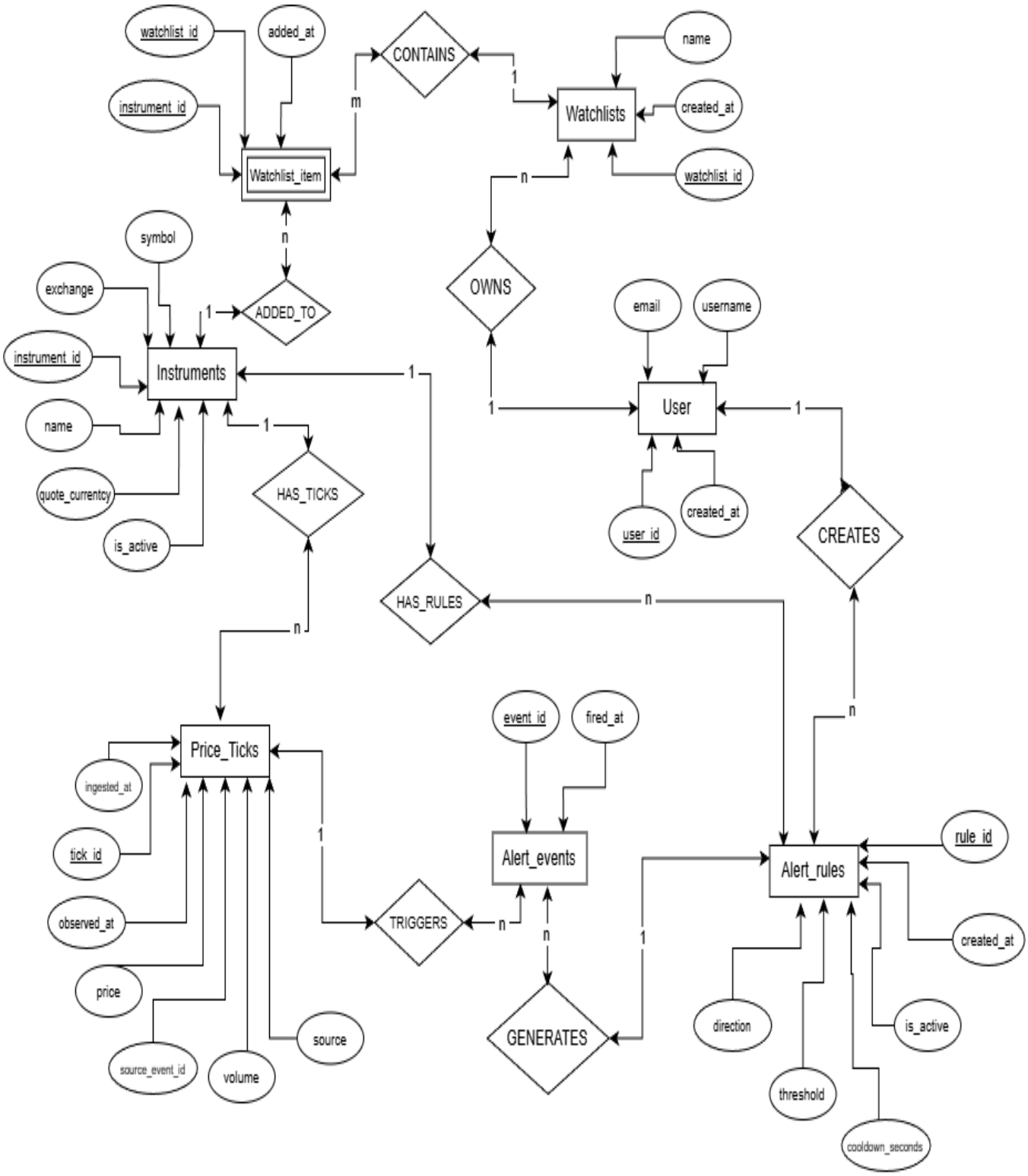
| Name | Registration Number | Phone Number/Email |
|---|---|---|
| Aditi Bajpai | 24BCE0372 | 8595997102 |
| Arshmeet Kaur Marhala | 24BCE0378 | 8360762414 |
| Yatharth Tongia | 24BCE0825 | 7619687973 |

# 4. PROJECT RESOURCE REQUIREMENTS

## 4.1 Software Resource Requirements

- Python 3.13
- PostgreSQL
- HTML, CSS
- JavaScript
- FastAPI
- Uvicorn

## 5. ER DIAGRAM


watchlist_id
added_at
CONTAINS
name
instrument_id
m
1
Watchlists
created_at
Watchlist_item
n
watchlist_id
symbol
n
exchange
OWNS
email
username
1
ADDED_TO
instrument_id
Instruments
1
1
User
1
name
1
quote_currency
HAS_TICKS
user_id
created_at
CREATES
is_active
HAS_RULES
n
n
n
event_id
fired_at
Price_Ticks
ingested_at
1
rule_id
Alert_events
Alert_rules
tick_id
observed_at
TRIGGERS
n
n
1
created_at
price
direction
is_active
source
GENERATES
source_event_id
volume
threshold
cooldown_seconds

## 6. Tables and Constraints

### 1. Table Name : users

| Attribute | Data Type | Constraint |
|---|---|---|
| user_id | BIGINT | PK, generated identity |
| name | TEXT | NOT NULL |
| email | TEXT | NOT NULL, UNIQUE |
| created_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

### 2. Table Name : instruments

| Attribute | Data Type | Constraint |
|---|---|---|
| instrument_id | BIGINT | PK, generated identity |
| symbol | TEXT | NOT NULL |
| name | TEXT | NOT NULL |
| exchange | TEXT | NOT NULL |
| currency | TEXT | NOT NULL |
| active | BOOLEAN | NOT NULL, default TRUE |
| created_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

### 3. Table Name : watchlists

| Attribute | Data Type | Constraint |
|---|---|---|
| watchlist_id | BIGINT | PK, generated identity |
| user_id | BIGINT | NOT NULL, FK → users |
| name | TEXT | NOT NULL |
| created_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

### 4. Table Name : watchlist_items

| Attribute | Data Type | Constraint |
|---|---|---|
| watchlist_id | BIGINT | PK/FK → watchlists |
| instrument_id | BIGINT | PK/FK → instruments |
| added_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

Composite Primary Key: (watchlist_id, instrument_id)

## 5. Table Name : price_ticks

| Attribute | Data Type | Constraint |
|---|---|---|
| tick_id | BIGINT | PK, generated identity |
| instrument_id | BIGINT | NOT NULL, FK → instruments |
| observed_at | TIMESTAMPTZ | NOT NULL |
| price | NUMERIC | NOT NULL, CHECK price > 0 |
| volume | NUMERIC | NOT NULL, CHECK volume ≥ 0 |
| source | TEXT | NOT NULL |
| source_event_id | TEXT | Used for source-event identity/idempotency |

## 6. Table Name : alert_rules

| Attribute | Data Type | Constraint |
|---|---|---|
| alert_rule_id | BIGINT | PK, generated identity |
| user_id | BIGINT | NOT NULL, FK → users |
| instrument_id | BIGINT | NOT NULL, FK → instruments |
| direction | TEXT | NOT NULL, CHECK: ABOVE/BELOW |
| threshold | NUMERIC | NOT NULL, positive |
| cooldown_seconds | INTEGER | NOT NULL, non-negative CHECK |
| enabled | BOOLEAN | NOT NULL, default TRUE |
| created_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

## 7. Table Name : alert_events

| Attribute | Data Type | Constraint |
|---|---|---|
| alert_event_id | BIGINT | PK, generated identity |
| alert_rule_id | BIGINT | NOT NULL, FK → alert_rules |
| tick_id | BIGINT | NOT NULL, FK → price_ticks |
| fired_at | TIMESTAMPTZ | NOT NULL, default current timestamp |

## 7. WORK BREAK DOWN

| Team Member Registration Number | Name | Work Assigned |
|---|---|---|
| 24BCE0372 | Aditi Bajpai | 1, 2, 4 |
| 24BCE0378 | Arshmeet Kaur Marhala | 1, 2, 6 |
| 24BCE0825 | Yatharth Tongia | 1, 2, 5 |

## 8. REVIEW EVALUATION

| COMPONENT | MARKS | Aditi | Arshmeet | Yatharth |
|---|---|---|---|---|
| ER-diagram | 3 | | | |
| Table Creation with description and screenshot | 5 | | | |
| Normal form | 2 | | | |
| TOTAL | 10 MARKS | | | |